

QCML Research

Unsupervised geometric signals that detect crises machine learning misses

Problem

Can we detect market regime shifts *without labeled crisis data*?

Approach

Embed time series into geometric space using spectral metric learning

Extract 3 geometric observables measuring manifold deformation during stress

Fully unsupervised — no crisis labels, no look-ahead

Outcome

Multi-Lag Fidelity (**d=0.56**) matches Random Forest rank — **without any labeled data**

Walk-forward: **86% detection**, 9-day delay across **12 crises** and **5 ETFs**

Solo-authored **25-page research paper** with 3 theorems

PythonPyTorchNumPy/SciPyOptunaLaTeX

Quanta Ventures

Developed systematic equity strategies for a quantitative fund

Problem

Building robust multi-strategy portfolios that survive regime changes

Approach

Modular framework: multiple strategy sleeves with volatility targeting

VIX/VVIX regime diagnostics for dynamic leverage

Walk-forward validation with 30-day embargo and robustness suite

2.92

Sharpe Ratio

5.02

Calmar Ratio

-0.070

Max Drawdown

133

Total Signals

Strategy Sleeves:

5 (Momentum, Mean Rev, Statistical, Risk Budget, Factor)

Validation:

Walk-forward with 30-day embargo + 11-test robustness suite

Outcome

Sharpe 2.92 and Calmar 5.02 on out-of-sample data

Passed full robustness test suite

Built C++ Heston pricer for options strategy component

Pythonscikit-learnOptunaPostgreSQLC++

Student Quant Fund

Finding alpha and building community

Problem

Giving underclassmen access to quantitative skills and a production trading platform

Approach

Teaching momentum and mean-variance strategies

Building production-ready platform for future cohorts

Preparing for paper trading and capital deployment

Outcome

Mentoring newer students in collaborative strategy development

Building institutional knowledge across cohorts

Platform designed for live paper trading

PythonJupyterTeam Collaboration

Skills

Hard Skills

Tech

Python (NumPy, Pandas, PyTorch)SQLGit/CI-CDockerLaTeXMathematica

CS

Data Structures & AlgorithmsML/PyTorchNumerical Optimization

Finance

Stochastic CalculusOptions Pricing (BS, Heston)Backtesting/Walk-ForwardRisk Metrics (Sharpe, VaR, CVaR)

Soft Skills

CS Teaching Assistant

Held weekly office hours, mentored 60+ students

Independent Researcher

Solo-authored 25-page research paper with 3 theorems

Team Collaboration

Student Quant Fund leadership

Technical Communication

Translating complex math for diverse audiences